

Experiment 8

Aim: To demonstrate the performance of KNN algorithm by using iris dataset.

Program

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

iris=load_iris()
data=pd.DataFrame(data=iris.data,columns=iris.feature_names)
data['Species']=iris.target
x=data.drop('Species',axis=1)
y=data['Species']

x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_state=1)
model=KNeighborsClassifier(n_neighbors=5)
model.fit(x_train,y_train)
y_pred=model.predict(x_test)
accuracy=accuracy_score(y_test,y_pred)
print("accuracy:",accuracy)
conf_matrix = confusion_matrix(y_test,y_pred)
plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='pink',
xticklabels=iris.target_names, yticklabels=iris.target_names)
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Confusion Matrix')
```

```
plt.show()
```

Output:

